

Introduction to Digital Logic

Lecture 4

Circuit Optimization

- **Two-Level Optimization**
 - **Map Manipulation**
- **Multi-Level Circuit Optimization**

Circuit Optimization

- **Goal: To obtain the simplest implementation for a given function**
- **Optimization is a more formal approach to simplification that is performed using a specific procedure or algorithm**
- **Optimization requires a cost criterion to measure the simplicity of a circuit**
- **Two distinct cost criteria we will use:**
 - **Literal cost (L)**
 - **Gate input cost (G)**
 - **Gate input cost with NOTs (GN)**

Literal Cost

- **Literal** – a variable or its complement
- **Literal cost** – the number of literal appearances in a Boolean expression corresponding to the logic circuit diagram

- **Examples:**

– $F = BD + A\bar{B}C + A\bar{C}\bar{D}$ $L = 8$

– $F = BD + A\bar{B}C + A\bar{B}\bar{D} + ABC\bar{C}$ $L =$

– $F = (A + B)(A + D)(B + C + \bar{D})(\bar{B} + \bar{C} + D)$ $L =$

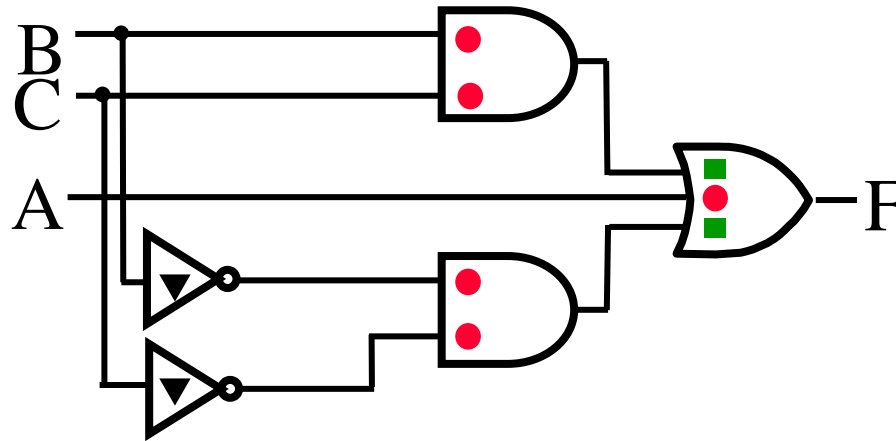
– Which solution is best?

Gate Input Cost

- Gate input costs - the number of inputs to the gates in the implementation corresponding exactly to the given equation or equations. (G - inverters not counted, GN - inverters counted)
- For SOP and POS equations, it can be found from the equation(s) by finding the sum of:
 - all literal appearances
 - the number of terms excluding terms consisting only of a single literal, (G) and
 - optionally, the number of distinct complemented single literals (GN).
- Example:
 - $F = BD + \overline{A}BC + A\overline{C}\overline{D}$ $G = 11, GN = 14$
 - $F = BD + \overline{A}BC + \overline{A}\overline{B}\overline{D} + ABC\overline{C}$ $G = \quad, GN =$
 - $F = (A + \overline{B})(A + D)(B + \overline{C} + D)(\overline{B} + \overline{C} + D)$ $G = \quad, GN =$
 - Which solution is best?

Cost Criteria (continued)

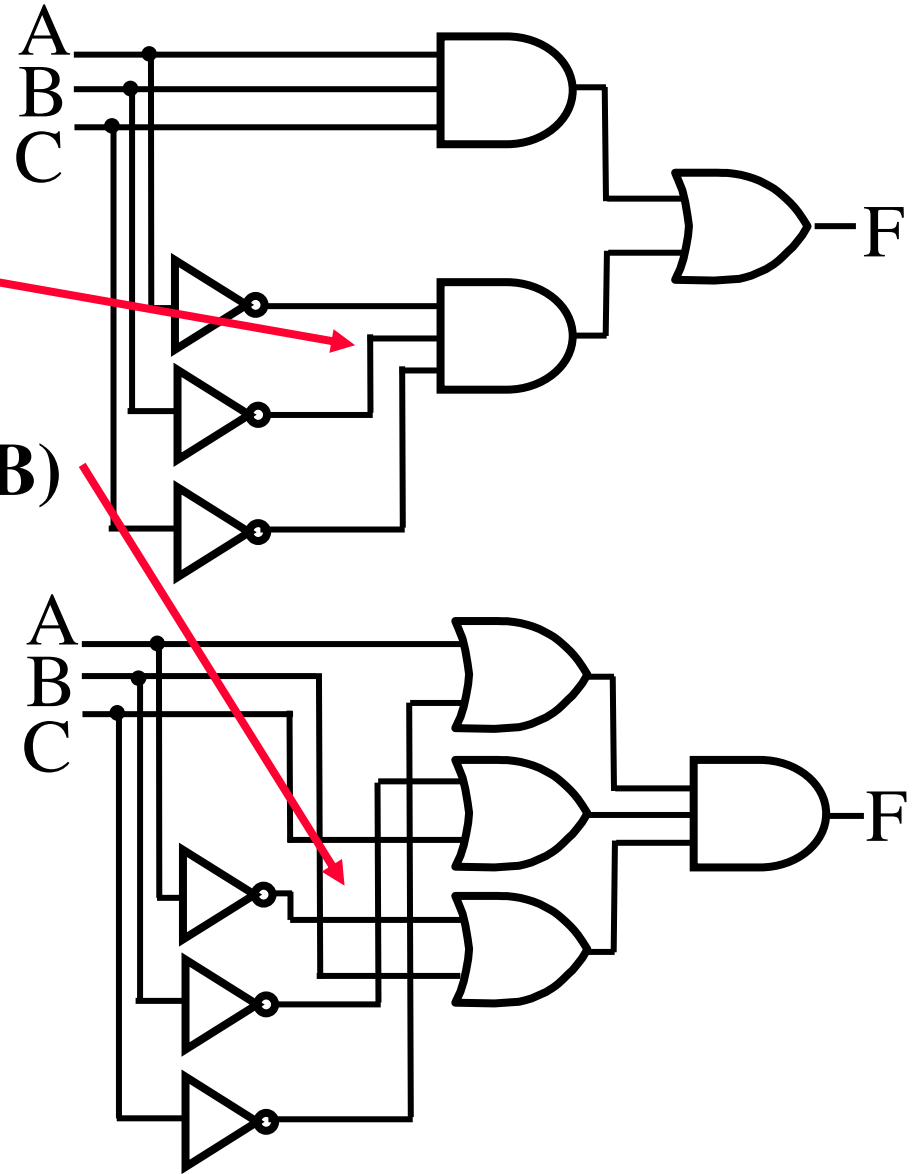
- Example 1: $\nabla \nabla$ $GN = G + 2 = 9$
- $F = A + B \underset{\text{■}}{C} + \bar{B} \underset{\text{■}}{\bar{C}}$ $L = 5$
- $G = L + 2 = 7$



- **L** (literal count) counts the AND inputs and the single literal OR input.
- **G** (gate input count) adds the remaining OR gate inputs
- **GN**(gate input count with NOTs) adds the inverter inputs

Cost Criteria (continued)

- **Example 2:**
- $F = A B C + \bar{A} \bar{B} \bar{C}$
- $L = 6 \quad G = 8 \quad GN = 11$
- $F = (A + \bar{C})(\bar{B} + C)(\bar{A} + B)$
- $L = 6 \quad G = 9 \quad GN = 12$
- Same function and same literal cost
- But first circuit has better gate input count and better gate input count with NOTs
- **Select it!**



Boolean Function Optimization

- Minimizing the gate input (or literal) cost of a (a set of) Boolean equation(s) reduces circuit cost.
- We choose gate input cost.
- Boolean Algebra and graphical techniques are tools to minimize cost criteria values.
- Some important questions:
 - When do we stop trying to reduce the cost?
 - Do we know when we have a minimum cost?
- Treat optimum or near-optimum cost functions for two-level (SOP and POS) circuits first.
- Introduce a graphical technique using Karnaugh maps (K-maps, for short)

Karnaugh Maps (K-map)

- **A K-map is a collection of squares**
 - Each square represents a minterm
 - The collection of squares is a graphical representation of a Boolean function
 - Adjacent squares differ in the value of one variable
 - Alternative algebraic expressions for the same function are derived by recognizing patterns of squares
- **The K-map can be viewed as**
 - A reorganized version of the truth table
 - A topologically-warped Venn diagram as used to visualize sets in algebra of sets

Some Uses of K-Maps

- **Provide a means for:**
 - **Finding optimum or near optimum**
 - **SOP and POS standard forms, and**
 - **two-level AND/OR and OR/AND circuit implementations**
 - for functions with small numbers of variables**
 - **Visualizing concepts related to manipulating Boolean expressions, and**
 - **Demonstrating concepts used by computer-aided design programs to simplify large circuits**

Two Variable Maps

- A 2-variable Karnaugh Map:

- Note that minterm m_0 and minterm m_1 are “adjacent” and differ in the value of the variable y

	$y = 0$	$y = 1$
$x = 0$	$\overline{m_0} = \overline{x} \overline{y}$	$\overline{m_1} = \overline{x} y$
$x = 1$	$m_2 = x \overline{y}$	$m_3 = x y$

- Similarly, minterm m_0 and minterm m_2 differ in the x variable.

- Also, m_1 and m_3 differ in the x variable as well.

- Finally, m_2 and m_3 differ in the value of the variable y

K-Map and Truth Tables

- The K-Map is just a different form of the truth table.
- Example – Two variable function:
 - We choose **a, b, c and d** from the set $\{0,1\}$ to implement a particular function, $F(x,y)$.

Function Table

Input Values (x,y)	Function Value $F(x,y)$
0 0	a
0 1	b
1 0	c
1 1	d

K-Map

	$y = 0$	$y = 1$
$x = 0$	a	b
$x = 1$	c	d

K-Map Function Representation

- **Example: $F(x,y) = x$**

F = x	y = 0	y = 1
x = 0	0	0
x = 1	1	1

- **For function $F(x,y)$, the two adjacent cells containing 1's can be combined using the Minimization Theorem:**

$$F(x,y) = x\bar{y} + xy = x$$

K-Map Function Representation

- **Example: $G(x,y) = x + y$**

$G = x+y$	$y = 0$	$y = 1$
$x = 0$	0	1
$x = 1$	1	1

- For $G(x,y)$, two pairs of adjacent cells containing 1's can be combined using the Minimization Theorem:

$$G(x, y) = (x \bar{y} + x y) + (x y + \bar{x} y) = x + y$$

Duplicate xy

Three Variable Maps

- A three-variable K-map:

	yz=00	yz=01	yz=11	yz=10
x=0	m ₀	m ₁	m ₃	m ₂
x=1	m ₄	m ₅	m ₇	m ₆

- Where each minterm corresponds to the product terms:

	yz=00	yz=01	yz=11	yz=10
x=0	$\bar{x} \bar{y} \bar{z}$	$\bar{x} \bar{y} z$	$\bar{x} y z$	$\bar{x} y \bar{z}$
x=1	$x \bar{y} \bar{z}$	$x \bar{y} z$	$x y z$	$x y \bar{z}$

- Note that if the binary value for an index differs in one bit position, the minterms are adjacent on the K-Map

Alternative Map Labeling

- Map use largely involves:
 - Entering values into the map, and
 - Reading off product terms from the map.
- Alternate labelings are useful:

	$\bar{y}$		y	
$\bar{x}$	0	1	3	2
x	4	5	7	6
	$\bar{z}$		z	

		$y \ z$			
		00	01	$\overbrace{11 \ 10}^y$	
x	0	0	1	3	2
	1	4	5	7	6
		$\underbrace{\hspace{1.5cm}}_z$			

Example Functions

- By convention, we represent the minterms of F by a "1" in the map and leave the minterms of $\bar{F}$ blank

- Example:

$$F(x, y, z) = \Sigma_m(2, 3, 4, 5)$$

			y	
	0	1	3_1	2_1
x	4_1	5_1	7	6

- Example:

$$G(a, b, c) = \Sigma_m(3, 4, 6, 7)$$

- Learn the locations of the 8 indices based on the variable order shown (x, most significant and z, least significant) on the map boundaries

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Combining Squares

- By combining squares, we reduce number of literals in a product term, reducing the literal cost, thereby reducing the other two cost criteria
- On a 3-variable K-Map:
 - One square represents a minterm with three variables
 - Two adjacent squares represent a product term with two variables
 - Four “adjacent” terms represent a product term with one variable
 - Eight “adjacent” terms is the function of all ones (no variables) = 1.

Example: Combining Squares

- Example: Let $F = \Sigma m(2,3,6,7)$

			y	
	0	1	3 1	2 1
x	4	5	7 1	6 1
			z	

- Applying the Minimization Theorem three times:

$$\begin{aligned}
 F(x, y, z) &= \bar{x} y z + x y z + \bar{x} y \bar{z} + x y \bar{z} \\
 &= yz + y\bar{z} \\
 &= y
 \end{aligned}$$

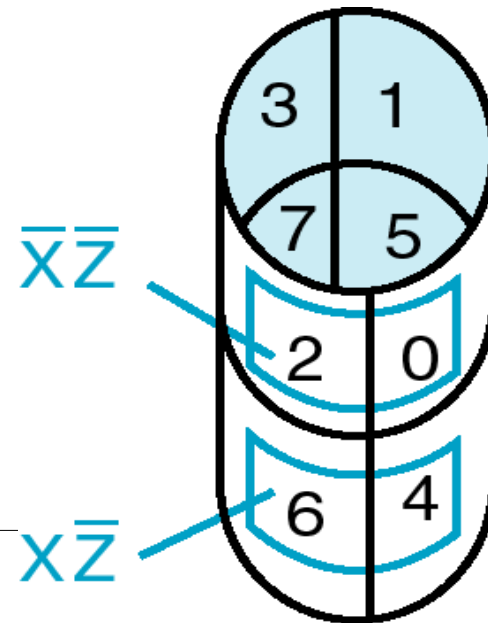
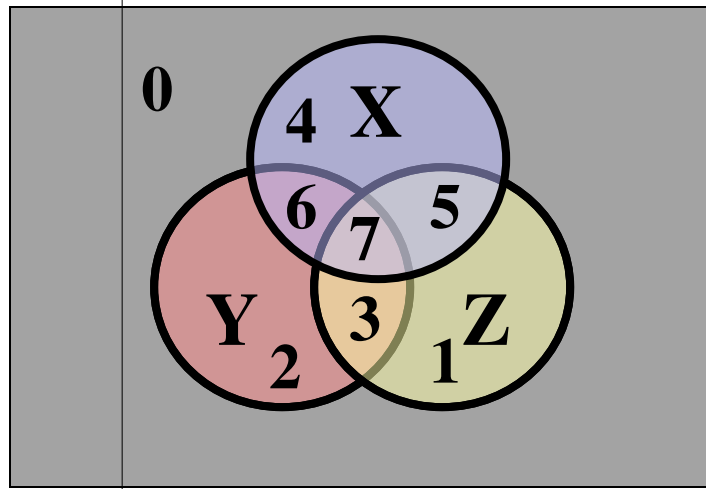
- Thus the four terms that form a 2×2 square correspond to the term "y".

Three-Variable Maps

- **Reduced literal product terms for SOP standard forms correspond to rectangles on K-maps containing cell counts that are powers of 2.**
- **Rectangles of 2 cells represent 2 adjacent minterms; of 4 cells represent 4 minterms that form a “pairwise adjacent” ring.**
- **Rectangles can contain non-adjacent cells as illustrated by the “pairwise adjacent” ring above.**

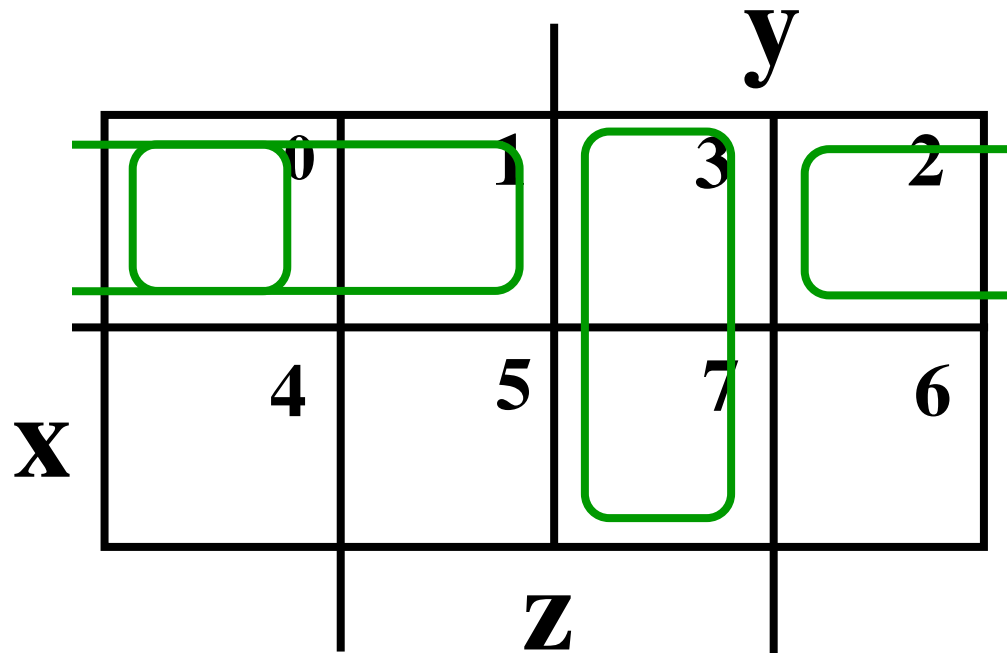
Three-Variable Maps

- Topological warps of 3-variable K-maps that show *all* adjacencies:
 - Venn Diagram
 - Cylinder



Three-Variable Maps

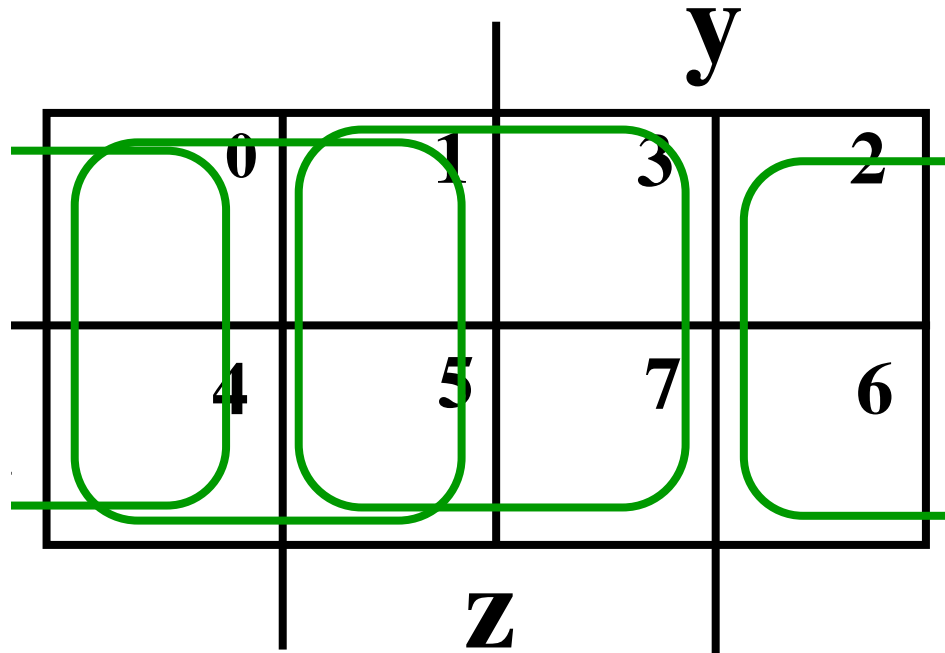
- **Example Shapes of 2-cell Rectangles:**



- **Read off the product terms for the rectangles shown**

Three-Variable Maps

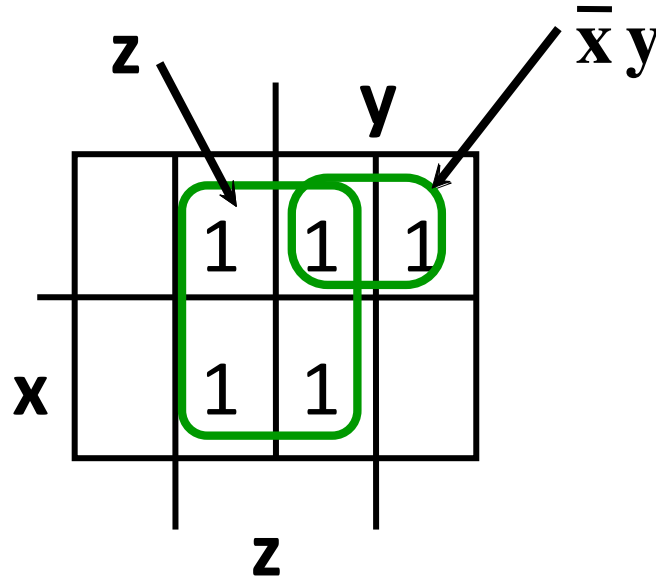
- **Example Shapes of 4-cell Rectangles:**



- **Read off the product terms for the rectangles shown**

Three Variable Maps

- K-Maps can be used to simplify Boolean functions by systematic methods. Terms are selected to cover the “1s” in the map.
- Example: Simplify $F(x, y, z) = \Sigma_m(1, 2, 3, 5, 7)$



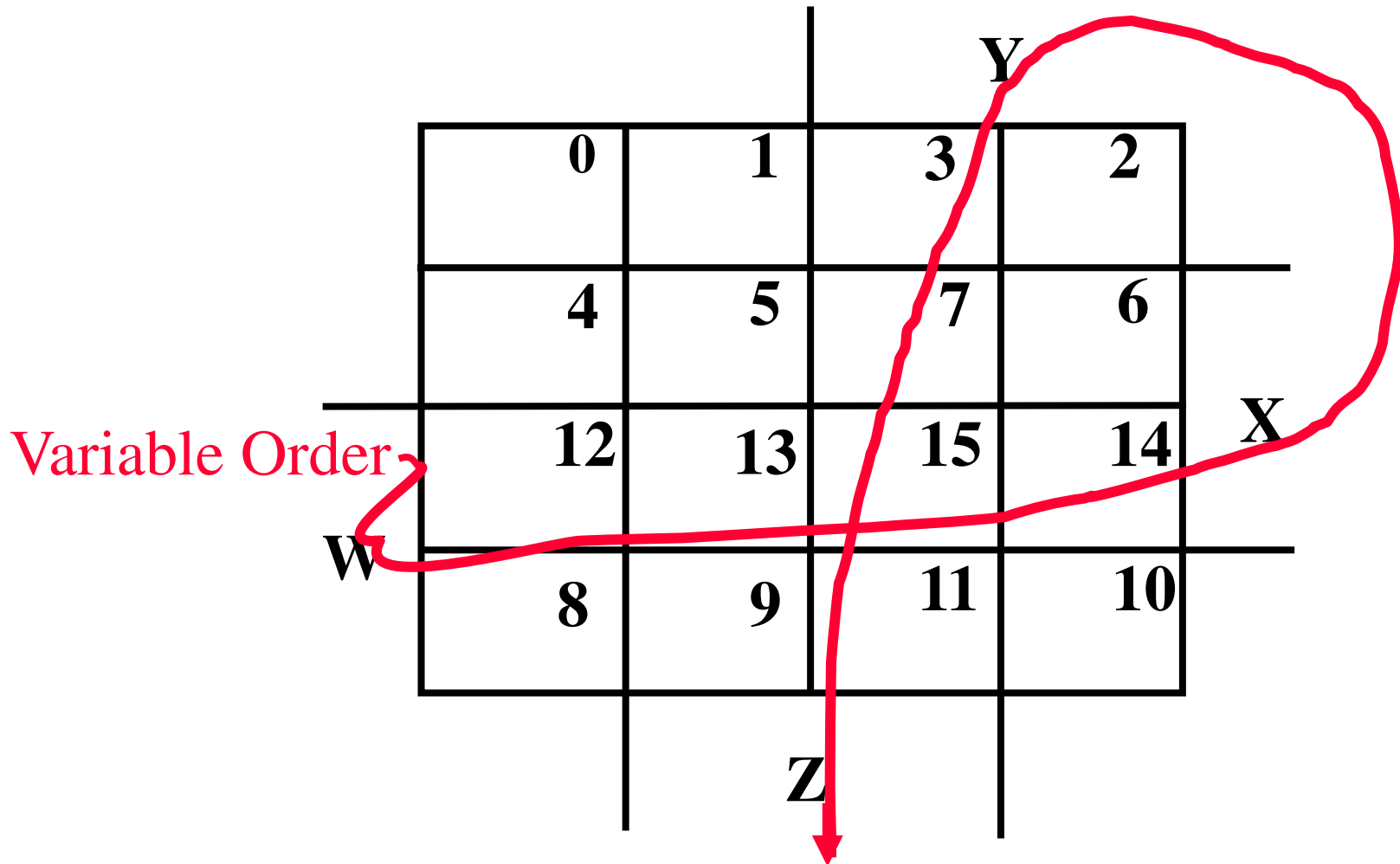
$$F(x, y, z) = z + \bar{x}y$$

Three-Variable Map Simplification

- Use a K-map to find an optimum SOP equation for $F(X, Y, Z) = \Sigma_m(0, 1, 2, 4, 6, 7)$

Four Variable Maps

- Map and location of minterms:

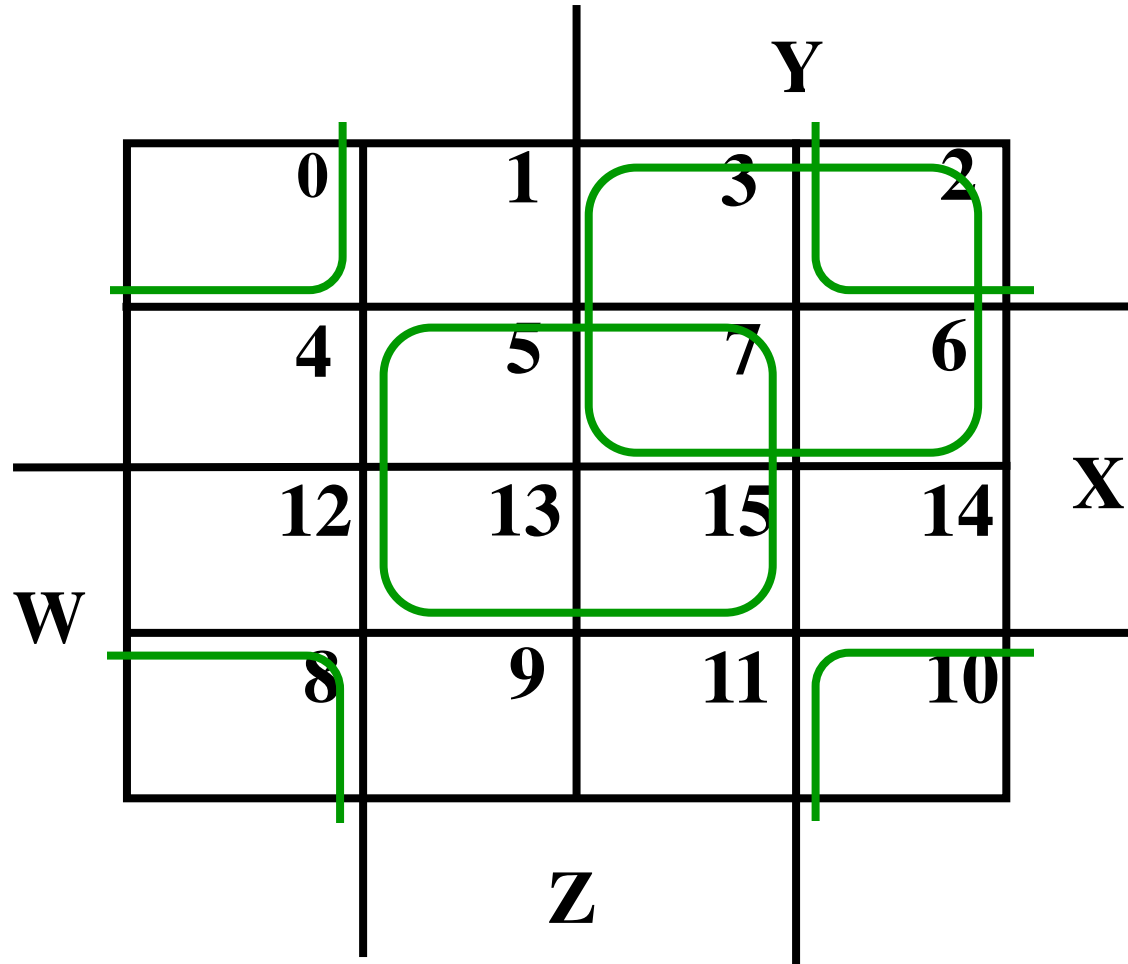


Four Variable Terms

- **Four variable maps can have rectangles corresponding to:**
 - **A single 1 = 4 variables, (i.e. Minterm)**
 - **Two 1s = 3 variables,**
 - **Four 1s = 2 variables**
 - **Eight 1s = 1 variable,**
 - **Sixteen 1s = zero variables (i.e. Constant "1")**

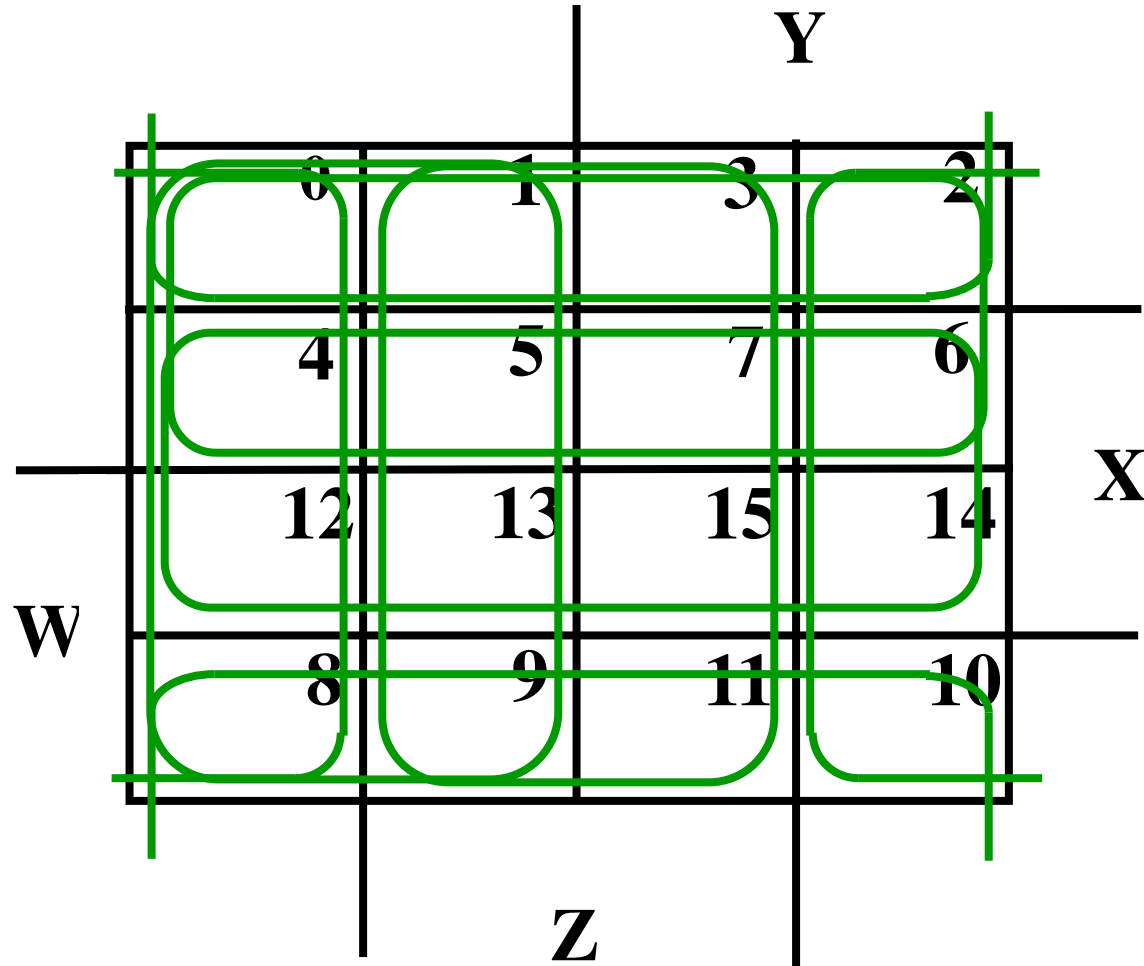
Four-Variable Maps

- **Example Shapes of Rectangles:**



Four-Variable Maps

- **Example Shapes of Rectangles:**



Four-Variable Map Simplification

- $F(W, X, Y, Z) = \Sigma_m(0, 2, 4, 5, 6, 7, 8, 10, 13, 15)$

Four-Variable Map Simplification

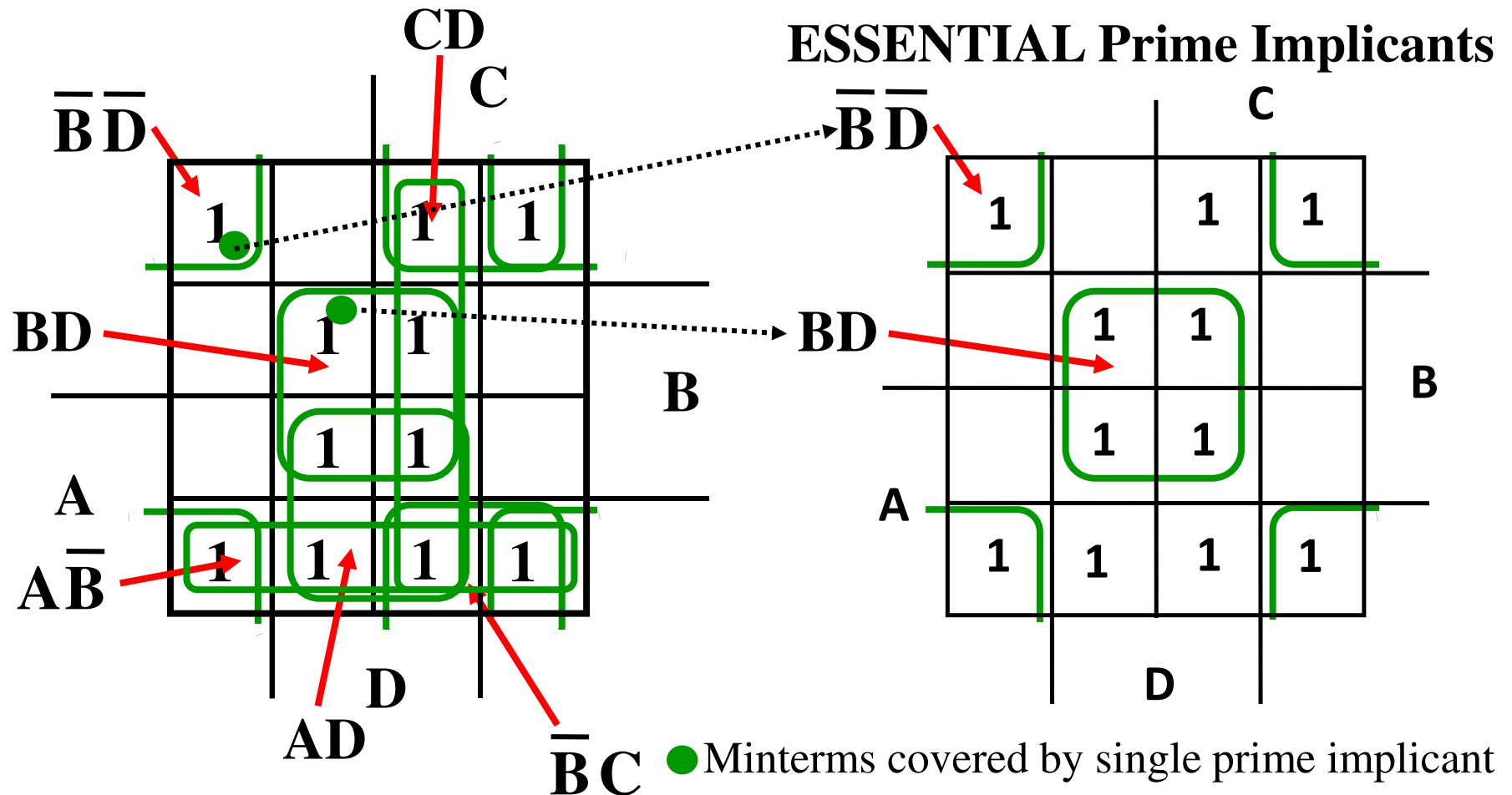
- $F(W, X, Y, Z) = \Sigma_m(3,4,5,7,9,13,14,15)$

Systematic Simplification

- A *Prime Implicant* is a product term obtained by combining the maximum possible number of adjacent squares in the map into a rectangle with the number of squares a power of 2.
- A prime implicant is called an *Essential Prime Implicant* if it is the only prime implicant that covers (includes) one or more minterms.
- Prime Implicants and Essential Prime Implicants can be determined by inspection of a K-Map.
- A set of prime implicants "*covers all minterms*" if, for each minterm of the function, at least one prime implicant in the set of prime implicants includes the minterm.

Example of Prime Implicants

- Find ALL Prime Implicants



Prime Implicant Practice

- Find all prime implicants for:
 $F(A,B,C,D) = \Sigma_m(0,2,3,8,9,10,11,12,13,14,15)$

Another Example

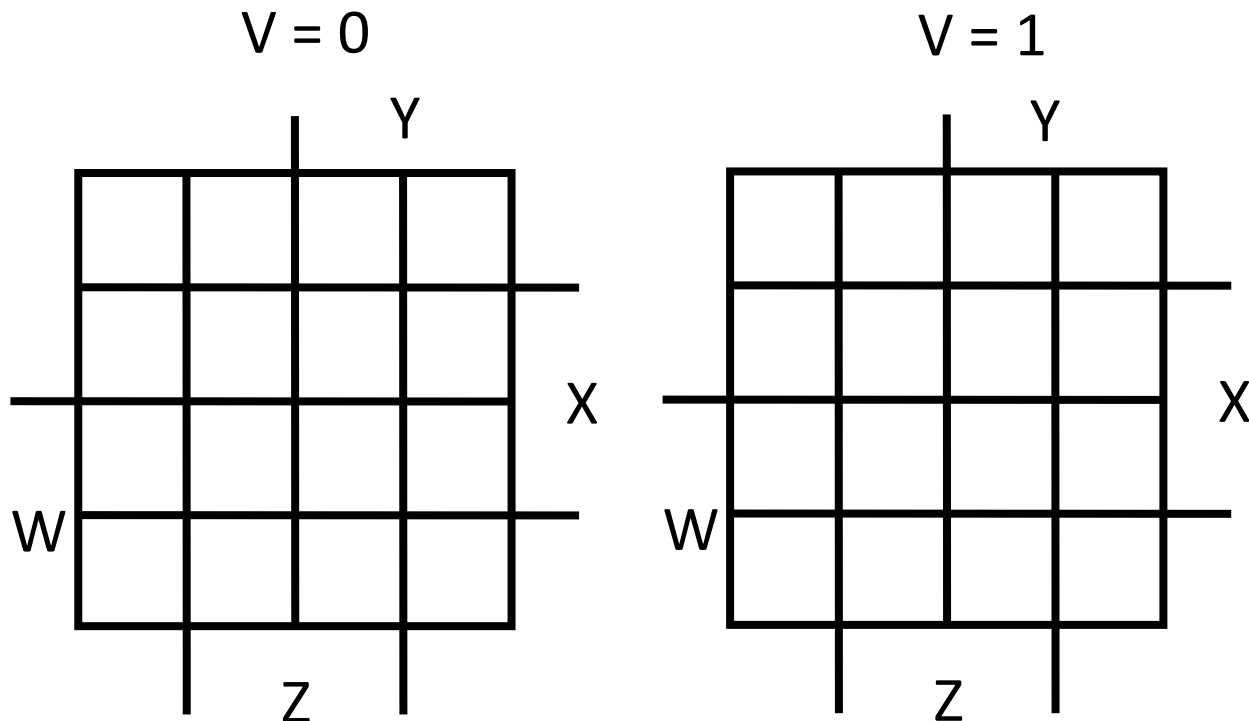
- Find all prime implicants for:

$$G(A,B,C,D) = \Sigma_m(0,2,3,4,7,12,13,14,15)$$

— Hint: There are seven prime implicants!

Five Variable or More K-Maps

- For five variable problems, we use *two adjacent K-maps*. It becomes harder to visualize adjacent minterms for selecting PIs. You can extend the problem to six variables by using four K-Maps.



Don't Cares in K-Maps

- Sometimes a function table or map contains entries for which it is known:
 - the input values for the minterm will never occur, or
 - The output value for the minterm is not used
- In these cases, the output value need not be defined
- Instead, the output value is defined as a “don't care”
- By placing “don't cares” (an “**x**” entry) in the function table or map, the cost of the logic circuit may be lowered.
- Example 1: A logic function having the binary codes for the BCD digits as its inputs. Only the codes for 0 through 9 are used. The six codes, 1010 through 1111 never occur, so the output values for these codes are “**x**” to represent “don't cares.”

Don't Cares in K-Maps

- **Example 2: A circuit that represents a very common situation that occurs in computer design has two distinct sets of input variables:**
 - **A, B, and C which take on all possible combinations, and**
 - **Y which takes on values 0 or 1.**

and a single output Z. The circuit that receives the output Z observes it only for $(A,B,C) = (1,1,1)$ and otherwise ignores it. Thus, Z is specified only for the combinations $(A,B,C,Y) = 1110$ and 1111 . For these two combinations, $Z = Y$. For all of the 14 remaining input combinations, Z is a don't care.

- Ultimately, each “**x**” entry may take on either a 0 or 1 value in resulting solutions
- For example, an “**x**” may take on value “0” in an SOP solution and value “1” in a POS solution, or vice-versa.
- Any minterm with value “**x**” need not be covered by a prime implicant.

Example: BCD “5 or More”

- The map below gives a function $F_1(w,x,y,z)$ which is defined as "5 or more" over BCD inputs. With the don't cares used for the 6 non-BCD combinations:

		y				
		0	1	3	2	
	0	0 ₄	1 ₅	1 ₇	1 ₆	
		X ₁₂	X ₁₃	X ₁₅	X ₁₄	x
w		1 ₈	1 ₉	X ₁₁	X ₁₀	
		z				

$$F_1(w,x,y,z) = w + xz + xy \quad G = 7$$

- This is much lower in cost than F_2 where the “don't cares” were treated as “0s.”

$$F_2(w, x, y, z) = \bar{w}xz + \bar{w}xy + w\bar{x}\bar{y} \quad G = 12$$

- For this particular function, cost G for the POS solution for $F_1(w,x,y,z)$ is not changed by using the don't cares.

Optimization Algorithm

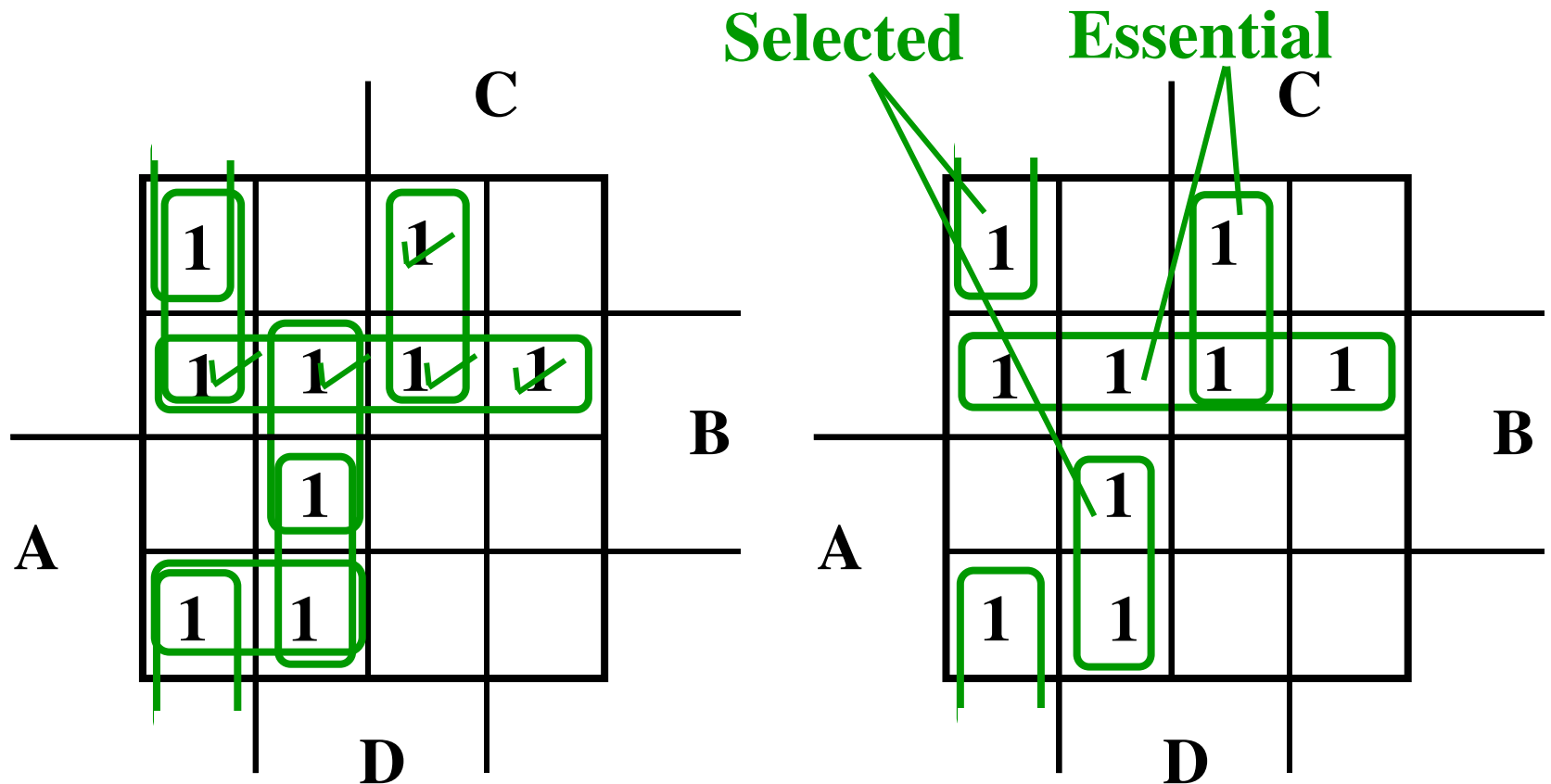
- Find all prime implicants.
- Include all essential prime implicants in the solution
- Select a minimum cost set of non-essential prime implicants to cover all minterms not yet covered:
 - Obtaining an optimum solution: See Reading Supplement - More on Optimization
 - Obtaining a good simplified solution: Use the Selection Rule

Prime Implicant Selection Rule

- **Minimize the overlap among prime implicants as much as possible. In particular, in the final solution, make sure that each prime implicant selected includes at least one minterm not included in any other prime implicant selected.**

Selection Rule Example

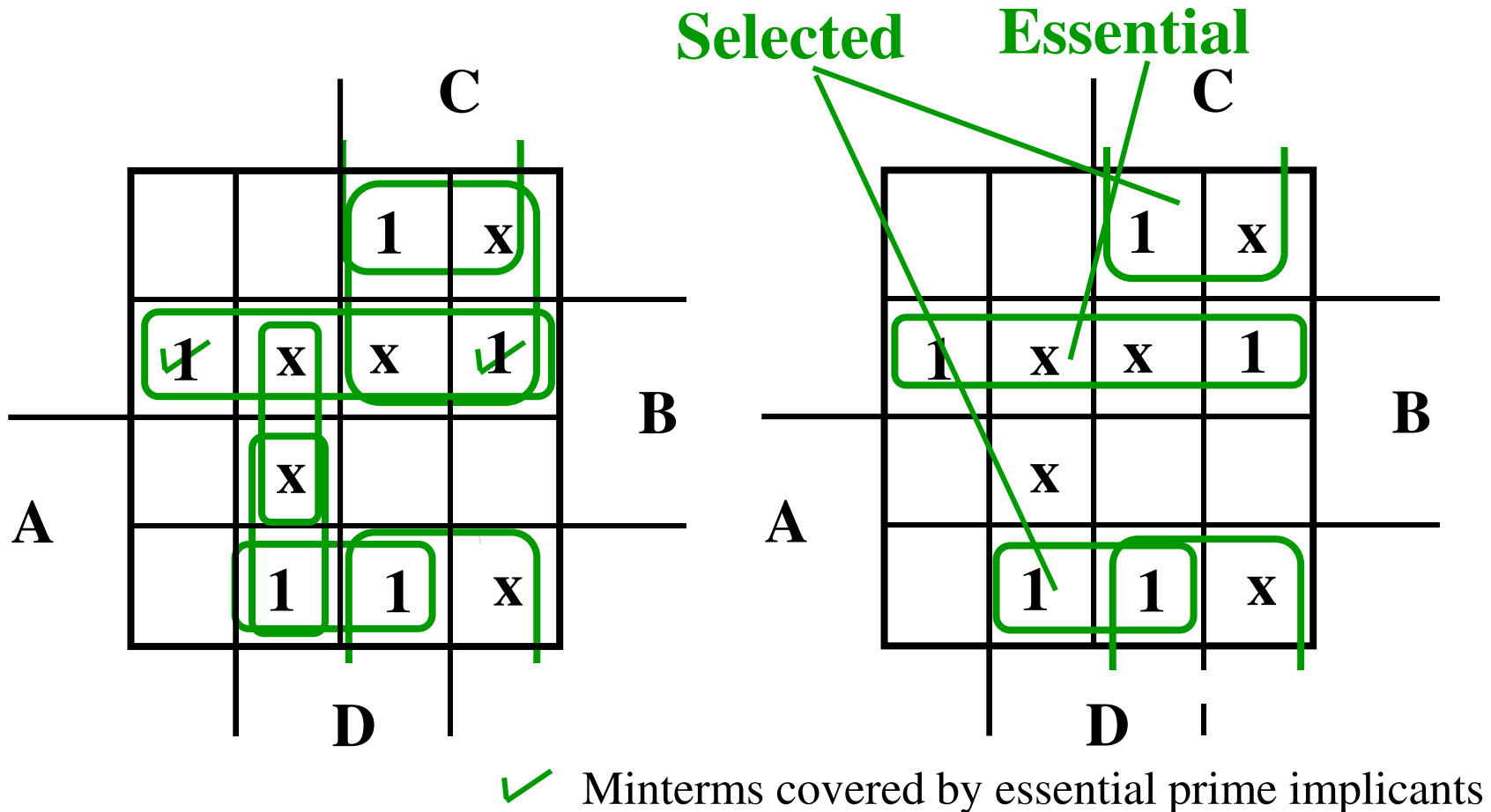
- Simplify $F(A, B, C, D)$ given on the K-map.



✓ Minterms covered by essential prime implicants

Selection Rule Example with Don't Cares

- Simplify $F(A, B, C, D)$ given on the K-map.



Multiple-Level Optimization

- **Multiple-level circuits - circuits that are not two-level (with or without input and/or output inverters)**
- **Multiple-level circuits can have reduced gate input cost compared to two-level (SOP and POS) circuits**
- **Multiple-level optimization is performed by applying transformations to circuits represented by equations while evaluating cost**

Transformations

- **Factoring - finding a factored form from SOP or POS expression**
 - Algebraic - No use of axioms specific to Boolean algebra such as complements or idempotence
 - Boolean - Uses axioms unique to Boolean algebra
- **Decomposition - expression of a function as a set of new functions**

Transformations (continued)

- **Substitution of G into F – expression of function F as a function of G and some or all of its original variables**
- **Elimination - Inverse of substitution**
- **Extraction - decomposition applied to multiple functions simultaneously**

Transformation Examples

- Algebraic Factoring

$$F = \bar{A} \bar{C} \bar{D} + \bar{A} B \bar{C} + ABC + AC \bar{D} \quad G = 16$$

– Factoring:

$$F = \bar{A} (\bar{C} \bar{D} + B \bar{C}) + A (BC + C \bar{D}) \quad G = 18$$

– Factoring again:

$$F = \bar{A} \bar{C} (B + \bar{D}) + AC (B + \bar{D}) \quad G = 12$$

– Factoring again:

$$F = (\bar{A} \bar{C} + AC) (B + \bar{D}) \quad G = 10$$

Transformation Examples

- **Decomposition**

- The terms $B + \bar{D}$ and $\bar{A}\bar{C} + AC$ can be defined as new functions E and H respectively, decomposing F :

$$F = E H, E = B + \bar{D}, \text{ and } H = \bar{A}\bar{C} + AC \quad G = 10$$

- This series of transformations has reduced G from 16 to 10, a substantial savings. The resulting circuit has three levels plus input inverters.

Transformation Examples

- Substitution of E into F

- Returning to F just before the final factoring step:

$$F = \bar{A} \bar{C} (B + \bar{D}) + AC (B + \bar{D}) \quad G = 12$$

- Defining $E = B + \bar{D}$, and substituting in F:

$$F = \bar{A} \bar{C} E + ACE \quad G = 10$$

- This substitution has resulted in the same cost as the decomposition

Transformation Examples

- **Elimination**

- **Beginning with a new set of functions:**

$$X = B + C$$

$$Y = A + B$$

$$Z = \bar{A}X + CY \qquad G = 10$$

- **Eliminating X and Y from Z:**

$$Z = \bar{A}(B + C) + C(A + B) \qquad G = 10$$

- **“Flattening” (Converting to SOP expression):**

$$Z = \bar{A}B + \bar{A}C + AC + BC \qquad G = 12$$

- **This has increased the cost, but has provided a new SOP expression for two-level optimization.**

Transformation Examples

- **Two-level Optimization**

- **The result of 2-level optimization is:**

$$Z = \bar{A}B + C$$

$$G = 4$$

- **This example illustrates that:**

- **Optimization can begin with any set of equations, not just with minterms or a truth table**
 - **Increasing gate input count G temporarily during a series of transformations can result in a final solution with a smaller G**

Transformation Examples

- **Extraction**

- **Beginning with two functions:**

$$E = \bar{A} \bar{B} \bar{D} + \bar{A} B D$$

$$H = \bar{B} C \bar{D} + B C D$$

$$G = 16$$

- **Finding a common factor and defining it as a function:**

$$F = \bar{B} \bar{D} + B D$$

- **We perform extraction by expressing E and H as the three functions:**

$$F = \bar{B} \bar{D} + B D, \quad E = \bar{A} F, \quad H = C F$$

$$G = 10$$

- **The reduced cost G results from the sharing of logic between the two output functions**